

# EAB 770 - Chapter 4

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## 1 $2 \times r$ and $s \times 2$ tables

### 1.1 Outline

- $2 \times r$  table
- Sets of  $2 \times r$  tables
- $s \times 2$  table
- Sets of  $s \times 2$  tables
  - Relationships between Sets of Tables
  - Exact Analysis of Association ( $s \times 2$ ) tables)

## 2 what if there are more than two outcomes?

### 2.1 Multiple Categorical Outcomes

- Likert Scale (Strongly Disagree, Disagree, Neutral, Agree, Strongly Agree)
- Part of a whole (None, Some, All)
- We would need more than two columns in our contingency table to account for these other outcomes.
  - Variables can be forced to be “binary”, but some important information can be lost.
- If there are  $r$  outcome levels and 2 levels of the exploratory variable, we use a  $2 \times r$  contingency table.
  - Focus on outcomes being ordinally scaled.

## 2.2 $2 \times r$ table

Let  $A_j$  be the  $j^{th}$  outcome level.

- Since  $A_j$  is ordinal, then  $A_1 < A_2 < A_3 < \dots A_r$ .
  - The  $\leq$  can replace  $<$  as long as at least one is a strict inequality

To account for ordinality, we assign a set of scores  $\mathbf{a}$  reflecting the relationship between response levels.

$$\mathbf{a} = \{a_1, a_2, a_3, \dots a_r\} : a_1 < a_2 < \dots < a_r$$

|       | $A_1$    | $A_2$    | ... | $A_r$    | Total    |
|-------|----------|----------|-----|----------|----------|
| $X_1$ | $n_{11}$ | $n_{12}$ | ... | $n_{1r}$ | $n_{1+}$ |
| $X_2$ | $n_{21}$ | $n_{22}$ | ... | $n_{2r}$ | $n_{2+}$ |
| Total | $n_{+1}$ | $n_{+2}$ | ... | $n_{+r}$ | $n$      |

## 2.3 Row Mean Scores

We can define a group mean for  $X_1$ , denoted by  $\bar{f}_1$ .

$$\bar{f}_1 = \sum_{j=1}^r \frac{a_j n_{1j}}{n_{1+}}$$

The random variable  $\bar{f}_1$  approximately has a normal distribution when the central limit theorem is imposed.

- High row sum

We refer to these scores as **Row Mean Scores**

## 2.4 2 × r table: Null hypothesis

Null hypothesis  $H_0$ : There is no association between the row (X) and column (A) variables.

Under the null hypothesis:

- The expected value of the mean can be calculated as:

$$\mu_a = \sum_{j=1}^r \frac{a_j n_{+j}}{n}$$

## 2.5 Mean Score Statistic

We can define a mean score statistic,  $Q_S$ , given by

$$Q_S = \frac{(\bar{f}_1 - \mu_1)^2}{V}$$

### **i** Note

- $V$  is the variance of  $\bar{f}_1$  under the null hypothesis.
- $Q_S$  follows a chi-square distribution with one degree of freedom.

## 2.6 Mean Score Statistic

- Targets alternative hypothesis of mean shifts to the hypothesis of no association
  - Uses fewer degrees of freedom compared to Pearson chi-square

### **i** Note

Pearson chi-square requires  $(r - 1)$  degrees of freedom compared to 1 degree of freedom for  $Q_S$

- The mean score statistic can also be used to determine the tendency of one group having better scores compared to the other.
- The  $Q_S$  sample size condition is less stringent:

### **i** Note

Choose one or more cutpoints  $j = (2, 3, \dots, (r-1))$ , add the first through  $j^{th}$  columns together and add the  $(j-1)^{th}$  through  $r^{th}$  columns together. If both of these sums are 5 or greater for at least one cutpoint in each row, then the sample size is adequate.

## 2.7 $2 \times r$ analysis: SAS

- Data should be encoded in increasing order of the outcome variable.
- The FREQ procedure will give us the mean score statistic:
  - **ORDER=DATA** should be included in the code with the PROC FREQ statement. (See sample code)
- Can be obtained using the cmh option or chisq option in the tables statement
  - $Q_S$ : Mantel-Haenszel Chi-square statistic in the SAS output
  - CMH: “Row Mean Scores Differ” statistic.
  - The CMH values would now display different values unlike the  $2 \times 2$  tables.

## 2.8 Example

The data shown in the table comes from a randomized, double-blind clinical trial investigating a new treatment for rheumatoid arthritis. Investigators compared the new treatment with a placebo; the response measured was whether there was no, some, or marked improvement in the symptoms of rheumatoid arthritis when the active treatment was administered.

- Is there sufficient evidence to conclude that the active treatment group has higher improvement compared to the placebo group?

**Table 4.1** Combined Rheumatoid Arthritis Data

| Treatment | Improvement |      |        | Total |
|-----------|-------------|------|--------|-------|
|           | None        | Some | Marked |       |
| Active    | 13          | 7    | 21     | 41    |
| Placebo   | 29          | 7    | 7      | 43    |
| Total     | 42          | 14   | 28     | 84    |

## 2.9 Example

The data shown in the table comes from a randomized, double-blind clinical trial investigating a new treatment for rheumatoid arthritis. Investigators compared the new treatment with a placebo; the response measured was whether there was no, some, or marked improvement in the symptoms of rheumatoid arthritis when the active treatment was administered.

- Is there sufficient evidence to conclude that the active treatment group has higher improvement compared to the placebo group?

### 2.9.1 SAS Code

```
data arth;
input treat $ response $ count @@;
datalines;
active none 13 active some 7 active marked 21
placebo none 29 placebo some 7 placebo marked 7
;
proc freq data=arth order=data;
weight count;
tables treat*response / cmh chisq nocol nopct;
run;
```

### 2.9.2 R Code

```
library(vcdExtra)
dat <- expand.grid(Treat=c("Active","Placebo"),
                  Response = c("None","Some","Marked")
                  ) |>
mutate(count = c(13,29,7,7,21,7))

xtabs(count ~ Treat+Response ,data=dat) -> dat_cc
dat_cc
```

|         | Response |      |        |
|---------|----------|------|--------|
| Treat   | None     | Some | Marked |
| Active  | 13       | 7    | 21     |
| Placebo | 29       | 7    | 7      |

```
CMHtest(dat_cc,
        types="ALL",
        overall=T)
```

Cochran-Mantel-Haenszel Statistics for Treat by Response

|         | AltHypothesis          | Chisq  | Df | Prob       |
|---------|------------------------|--------|----|------------|
| general | General association    | 12.900 | 2  | 0.00158084 |
| rmeans  | Row mean scores differ | 12.859 | 1  | 0.00033586 |
| cmeans  | Col mean scores differ | 12.900 | 2  | 0.00158084 |
| cor     | Nonzero correlation    | 12.859 | 1  | 0.00033586 |

### 2.9.3 Answer

$Q_S = 12.86$ ,  $p=0.0003$ . We have strong evidence to conclude that the active treatment displayed higher improvement compared to the placebo group.

- The active group has a higher proportion of marked + some improvement, which enables us to formulate the following conclusion.

### 2.10 Exercise

A survey was administered to students at a US-based university who are habitual smokers to determine the likelihood of using a mobile app to help them quit smoking. Is there sufficient evidence to conclude that heavy smokers are less likely to use a mobile app compared to light smokers?

|         | Response      |          |        |             |
|---------|---------------|----------|--------|-------------|
| Smokers | Very Unlikely | Unlikely | Likely | Very Likely |
| Heavy   | 36            | 75       | 56     | 17          |
| Light   | 38            | 59       | 64     | 23          |

### 2.11 Exercise

A survey was administered to students at a US-based university who are habitual smokers to determine the likelihood of using a mobile app to help them quit smoking. Is there sufficient evidence to conclude that heavy smokers are less likely to use a mobile app compared to light smokers?

|       | Very Unlikely | Unlikely | Likely | Very Likely |
|-------|---------------|----------|--------|-------------|
| Heavy | 36            | 75       | 56     | 17          |
| Light | 38            | 59       | 64     | 23          |

### 2.11.1 SAS Code

```
data smoke;
input treat $ response $ count @@;
datalines;
heavy vu 36 heavy u 75 heavy l 56 heavy vl 17
light vu 38 light u 59 light l 64 light vl 23
;

proc freq data=smoke order=data;
weight count;
tables treat*response / cmh chisq nocol nopct;
run;
```

### 2.11.2 R Code

```
library(vcdExtra)
CMHtest(dat_cc,
        types="rmeans",
        overall=T)
```

Cochran-Mantel-Haenszel Statistics for Smokers by Response

|        | AltHypothesis          | Chisq  | Df | Prob    |
|--------|------------------------|--------|----|---------|
| rmeans | Row mean scores differ | 1.0394 | 1  | 0.30795 |

### 2.11.3 Answer

$Q_S = 1.04$ ,  $p=0.31$ . There is insufficient evidence that heavy smokers are less likely to use a mobile app compared to light smokers.

### 3 Sets of $2 \times r$ tables

#### 3.1 Sets of $2 \times r$ tables

- We extend the Mantel-Haenszel strategy to  $2 \times r$  tables for ordinal responses
- Compute mean scores for the responses
  - Calculate mean score differences across tables
  - Similar approach to the difference in proportions ( $Q_S$ )
- Assume hypergeometric distributions for the contingency tables in the different strata.
  - Also assume that the across-strata sample sizes are sufficiently large for each group (row sums)

#### 3.2 Extended Mean Score Statistic

- $Q_{SMH}$  is the extended Mantel-Haenszel mean score statistic.
  - Also known as the ANOVA statistic
  - Function of the weighted average of the differences in mean scores of the two treatments for all strata.
- $Q_{SMH}$  is good for detecting consistent patterns of differences across strata.
- Sample size requirements involve collapsing each stratum table into a  $2 \times 2$  table by choosing cutpoints (Just like that in  $Q_S$ ), then apply the Mantel-Fleiss criterion.

#### 3.3 Different Types of Scores

##### 3.3.1 Integer Scores

- $(0, 1, 2, \dots)$
- Response levels are ordered categories that correspond to discrete counts.
- Produce same results regardless of starting number.
- Default in SAS.

##### 3.3.2 Rank

- Rank scores based on the row totals.
- In stratified analysis, rank scores are not robust when there are large strata.

$$R_i^1 = \sum_{k < i} n_{k+} + \frac{n_{i+} + 1}{2}$$

### Redit - Redit scores are standardized rank scores based on the sample size

$$R_i^3 = 1/(n + 1) \left[ \sum_{k < i} n_{k+} + \frac{n_{i+} + 1}{2} \right]$$

### 3.3.3 Standardized Midranks

- Also known as modified ridit scores.
- Constrained to lie between 0 and 1.
- Requires no scaling of response levels.
- Used when there is no argument for equal spaces between ordinal levels

$$R_i^3 = 1/(n + 1) \left[ \sum_{k < i} n_{k+} + \frac{n_{i+} + 1}{2} \right]$$

### 3.3.4 Logrank Scores

- Useful when the distribution is L-shaped.
- Need to apply the logarithmic transformation of the rank scores before applying to the `tables` statement.

## 3.4 Different Types of Scores

#### **i** Note

When there is no strata, the rank, ridit, and modridit scores correspond to the Wilcoxon rank-sum test.

#### **i** Note

In stratified analyses, the *modridit scores* are preferred out of these rank-based scores because it produces an extension of the Wilcoxon rank-sum test.

### 3.5 Measures of Association in Sets of $2 \times r$ tables

#### 3.5.1 Somers' D

- Measure of association for ordinally scaled data.
- For ordinal scores with two groups, let  $G1$  be the score of group 1 and  $G2$  be the score of group 2

$$D = \frac{P(G1 > G2) - P(G1 < G2)}{P(G1 > G2) + P(G1 < G2) + P(G1 = G2)}$$

#### Note

SAS will provide two values of the Somers'D:  $D(R|C)$  and  $D(C|R)$

- $D(R|C)$  treats the *column* as the independent variable.
- $D(C|R)$  treats the *row* as the independent variable

#### 3.5.2 Mann-Whitney rank measure of association

- Assesses association between a dichotomous predictor variable and an ordinal outcome.
- Can be calculated from Somers'D

$$g_h = (D + 1)/2$$

where the standard error of  $g_h$  is  $v_h = v_D/2$ .

### 3.6 Extension of the Mann-Whitney Rank Measure

- When there are  $q$  strata, we can calculate a test statistic  $Q_H$  such that

$$Q_H = \sum_{h=1}^q (g_h - \tilde{g})^2 / v_h$$
$$\tilde{g} = \frac{\sum_{h=1}^q g_h (n_{h1}n_{h2}/(n_{h1} + n_{h2}))}{\sum_{h=1}^q (n_{h1}n_{h2}/(n_{h1} + n_{h2}))}$$

- This is an extension of the van Elteren's rank sum test.

### 3.7 Scores: SAS

- The default score used in SAS are integer/table scores
- Scoring method can be specified using the SCORES option in SAS
  - TABLE: integer
  - RIDIT: midranks
  - MODRIDIT: standardized midranks
  - RANK: rank scores.
- Use the Cochran-Mantel-Haenszel table (CMH) in testing the hypothesis
  - Difference in mean scores: “Row means differ”

```
proc freq;  
tables A*B/scores=MODRIDIT;  
run;
```

### 3.8 Somers' D: SAS

- measures includes the Somers' D estimate with the standard errors (ASE column).
- The Mann-Whitney rank-sum statistic can be calculated using PROC IML.
- The cl option calculates confidence intervals for Somers' D

```
proc freq;  
tables A*B/scores=MODRIDIT measures cl;  
run;
```

### 3.9 Mann-Whitney: SAS (optional)

```
proc freq ;  
weight count;  
tables S*A*B/ measures scores=modridit;  
output out=somerout smdcr;  
ods output CrosstabFreqs=myFreqs;  
run;  
data ns;  
set myFreqs; where _type_="110";  
run;  
  
proc iml;
```

①  
②

```

use somerout var{_SMDCR_ E_SMDCR};
read all into combined;
use ns var{Frequency};
read all into total;
print total;
WH=(total[1,]#total[2,]) / (total[1,]+total[2,]) //
(total[3,]#total[4,]) / (total[3,]+total[4,]);
print wh;
GH=(combined[,1]+1)/2;
SEGH=(combined[,2])/2;
print wh GH SEGH;
QH= (GH[1] - GH[2])**2 / SSQ(SEGH);
pvalue=1-probchi(QH,1);
print GH SEGH QH pvalue;
gtilde1=sum(GH # WH);
gtilde2=sum(WH);
gtilde=gtilde1/gtilde2;
gtildevar=(sum(WH##2 # SEGH##2)) / ((sum(WH))**2);
segtilde=sqrt(gtildevar);
qgtilde=(gtilde-.5)**2/gtildevar;
print gtilde gtildevar qgtilde;
quit;

```

- ① Outputs somers'D(C|R) and error
- ② crosstabs outputs the contingency table

### 3.10 Example

The data shown in the table comes from a randomized, double-blind clinical trial investigating a new treatment for rheumatoid arthritis. Investigators compared the new treatment with a placebo; the response measured was whether there was no, some, or marked improvement in the symptoms of rheumatoid arthritis when the active treatment was administered.

- Is there sufficient evidence to conclude that the active treatment group has higher improvement compared to the placebo group after controlling for gender?
- Compare the results using table scores and modridit scores.

**Table 4.2** Rheumatoid Arthritis Data

| Gender | Treatment | Improvement |      |        | Total |
|--------|-----------|-------------|------|--------|-------|
|        |           | None        | Some | Marked |       |
| Female | Active    | 6           | 5    | 16     | 27    |
| Female | Placebo   | 19          | 7    | 6      | 32    |
| Total  |           | 25          | 12   | 22     | 59    |
| Male   | Active    | 7           | 2    | 5      | 14    |
| Male   | Placebo   | 10          | 0    | 1      | 11    |
| Total  |           | 17          | 2    | 6      | 25    |

### 3.11 Example

- Is there sufficient evidence to conclude that the active treatment group has higher improvement compared to the placebo group after controlling for gender?
- Compare the results using table scores and modridit scores.

**Table 4.2** Rheumatoid Arthritis Data

| Gender | Treatment | Improvement |      |        | Total |
|--------|-----------|-------------|------|--------|-------|
|        |           | None        | Some | Marked |       |
| Female | Active    | 6           | 5    | 16     | 27    |
| Female | Placebo   | 19          | 7    | 6      | 32    |
| Total  |           | 25          | 12   | 22     | 59    |
| Male   | Active    | 7           | 2    | 5      | 14    |
| Male   | Placebo   | 10          | 0    | 1      | 11    |
| Total  |           | 17          | 2    | 6      | 25    |

#### 3.11.1 SAS Code

```
data arth;  
input gender $ treat $ response $ count @@;  
datalines;
```

```

female active none 6 female active some 5 female active marked 16
female placebo none 19 female placebo some 7 female placebo marked 6
male active none 7 male active some 2 male active marked 5
male placebo none 10 male placebo some 0 male placebo marked 1
;

proc freq data=arth order=data;
weight count;
tables gender*treat*response / cmh nocol nopct;
tables gender*treat*response / cmh nocol nopct scores=modridit;
run;

```

- ① Table Scores
- ② Modridit scores (Standardized Midranks)

### 3.11.2 R Code

```

dat <- expand.grid(Improvement=c("None","Some","Marked"),
                  Treatment = c("Active","Placebo"),
                  gender=c("Female","Male")
                  ) |>
mutate(count = c(6,5,16,19,7,6,7,2,5,10,0,1))
xtabs(count ~ Treatment + Improvement + gender ,data=dat) -> dat_cc
dat_cc
CMHtest(dat_cc,
        types="ALL",
        cscores = 1:ncol(dat_cc),
        overall=T, strata="gender")

CMHtest(dat_cc,
        types="ALL",
        overall=T,
        cscores="midrank",
        strata="gender")

```

- ① table scores
- ② standardized midrank scores

, , gender = Female

|           | Improvement |      |        |
|-----------|-------------|------|--------|
| Treatment | None        | Some | Marked |
| Active    | 6           | 5    | 16     |
| Placebo   | 19          | 7    | 6      |

, , gender = Male

|           | Improvement |      |        |
|-----------|-------------|------|--------|
| Treatment | None        | Some | Marked |
| Active    | 7           | 2    | 5      |
| Placebo   | 10          | 0    | 1      |

\$`gender:Female`

Cochran-Mantel-Haenszel Statistics for Treatment by Improvement  
in stratum gender:Female

|         | AltHypothesis          | Chisq  | Df | Prob       |
|---------|------------------------|--------|----|------------|
| general | General association    | 11.105 | 2  | 0.00387828 |
| rmeans  | Row mean scores differ | 10.935 | 1  | 0.00094361 |
| cmeans  | Col mean scores differ | 11.105 | 2  | 0.00387828 |
| cor     | Nonzero correlation    | 10.935 | 1  | 0.00094361 |

\$`gender:Male`

Cochran-Mantel-Haenszel Statistics for Treatment by Improvement  
in stratum gender:Male

|         | AltHypothesis          | Chisq  | Df | Prob     |
|---------|------------------------|--------|----|----------|
| general | General association    | 4.7105 | 2  | 0.094871 |
| rmeans  | Row mean scores differ | 3.7128 | 1  | 0.053997 |
| cmeans  | Col mean scores differ | 4.7105 | 2  | 0.094871 |
| cor     | Nonzero correlation    | 3.7128 | 1  | 0.053997 |

\$ALL

Cochran-Mantel-Haenszel Statistics for Treatment by Improvement  
Overall tests, controlling for all strata

|         | AltHypothesis          | Chisq  | Df | Prob       |
|---------|------------------------|--------|----|------------|
| general | General association    | 14.632 | 2  | 0.00066473 |
| rmeans  | Row mean scores differ | 14.632 | 1  | 0.00013068 |
| cmeans  | Col mean scores differ | 14.632 | 2  | 0.00066473 |
| cor     | Nonzero correlation    | 14.632 | 1  | 0.00013068 |

```
$`gender:Female`
```

```
Cochran-Mantel-Haenszel Statistics for Treatment by Improvement  
in stratum gender:Female
```

|         | AltHypothesis          | Chisq  | Df | Prob       |
|---------|------------------------|--------|----|------------|
| general | General association    | 11.105 | 2  | 0.00387828 |
| rmeans  | Row mean scores differ | 10.879 | 1  | 0.00097249 |
| cmeans  | Col mean scores differ | 11.105 | 2  | 0.00387828 |
| cor     | Nonzero correlation    | 10.879 | 1  | 0.00097249 |

```
$`gender:Male`
```

```
Cochran-Mantel-Haenszel Statistics for Treatment by Improvement  
in stratum gender:Male
```

|         | AltHypothesis          | Chisq  | Df | Prob     |
|---------|------------------------|--------|----|----------|
| general | General association    | 4.7105 | 2  | 0.094871 |
| rmeans  | Row mean scores differ | 4.1469 | 1  | 0.041712 |
| cmeans  | Col mean scores differ | 4.7105 | 2  | 0.094871 |
| cor     | Nonzero correlation    | 4.1469 | 1  | 0.041712 |

```
$ALL
```

```
Cochran-Mantel-Haenszel Statistics for Treatment by Improvement  
Overall tests, controlling for all strata
```

|         | AltHypothesis          | Chisq  | Df | Prob       |
|---------|------------------------|--------|----|------------|
| general | General association    | 14.632 | 2  | 0.00066473 |
| rmeans  | Row mean scores differ | 15.004 | 1  | 0.00010728 |
| cmeans  | Col mean scores differ | 14.632 | 2  | 0.00066473 |
| cor     | Nonzero correlation    | 15.004 | 1  | 0.00010728 |

### 3.11.3 Answer

$Q_{SMH} = 14.6$ ,  $p < 0.001$ . There is sufficient evidence to conclude that the active treatment group has higher improvement compared to the placebo group.

When using the modridit scores,  $Q_{SMH} = 15.00$ ,  $p < 0.001$ . Same conclusion with a slight deviation between statistics.

### 3.12 Exercise

The following data is obtained from the 2003 Programme for International Student Assessment (PISA) study.

- Use the appropriate analysis to determine whether the attitude toward school is different between students in the US and Korea after controlling for gender.
- Calculate the relevant Somers' D values for each gender.

#### 3.12.1 Contingency Table

| Country       | Gender  | To what extent do you agree that school has been a waste of time? |       |          |                   |
|---------------|---------|-------------------------------------------------------------------|-------|----------|-------------------|
|               |         | Strongly agree                                                    | Agree | Disagree | Strongly disagree |
| United States | Females | 38                                                                | 125   | 1227     | 1253              |
|               | Males   | 102                                                               | 249   | 1369     | 922               |
| Korea         | Females | 26                                                                | 199   | 1415     | 578               |
|               | Males   | 58                                                                | 281   | 1942     | 897               |

#### 3.12.2 SAS Data

```
data school;
input gender $ location $ response $ count @@;
datalines;
female US sd 1253 female US d 1227 female US a 125 female US sa 38
female korea sd 578 female korea d 1415 female korea a 199 female korea sa 26
male US sd 922 male US d 1369 male US a 249 male US sa 102
male korea sd 897 male korea d 1942 male korea a 281 male korea sa 58
;
```

### 3.13 Exercise

The following data is obtained from the 2003 Programme for International Student Assessment (PISA) study.

- Use the appropriate analysis to determine whether the attitude toward school is different between students in the US and Korea after controlling for gender.
- Calculate the relevant Somers' D values for each gender.

### 3.13.1 SAS Code

```
proc freq data=school order=data;
weight count;
tables gender*location*response/all nocol nopct measures cl;
tables gender*location*response/all nocol nopct measures scores=modridit cl;
run;
```

### 3.13.2 R Code

```
dat <- expand.grid(Rating=c("StD","D","A","StA"),
                  gender=c("Female","Male"),
                  country = c("US","Korea")) |>
mutate(count = c(1253,1227,125,38,922,1369,249,102,578,1415,199,26,897,1942,281,58))

xtabs(count ~ country +Rating + gender ,data=dat) -> dat_cc
dat_cc
```

```
, , gender = Female
```

|         | Rating |      |     |     |
|---------|--------|------|-----|-----|
| country | StD    | D    | A   | StA |
| US      | 1253   | 1227 | 125 | 38  |
| Korea   | 578    | 1415 | 199 | 26  |

```
, , gender = Male
```

|         | Rating |      |     |     |
|---------|--------|------|-----|-----|
| country | StD    | D    | A   | StA |
| US      | 922    | 1369 | 249 | 102 |
| Korea   | 897    | 1942 | 281 | 58  |

```
CMHtest(dat_cc,
        types="ALL",
        cscores = 1:ncol(dat_cc),
        overall=T, strata="gender")
```

```
$`gender:Female`
```

```
Cochran-Mantel-Haenszel Statistics for country by Rating
```

```
in stratum gender:Female
```

|         | AltHypothesis          | Chisq  | Df | Prob       |
|---------|------------------------|--------|----|------------|
| general | General association    | 246.04 | 3  | 4.6999e-53 |
| rmeans  | Row mean scores differ | 182.24 | 1  | 1.5754e-41 |
| cmeans  | Col mean scores differ | 246.04 | 3  | 4.6999e-53 |
| cor     | Nonzero correlation    | 182.24 | 1  | 1.5754e-41 |

```
$`gender:Male`
```

```
Cochran-Mantel-Haenszel Statistics for country by Rating  
in stratum gender:Male
```

|         | AltHypothesis          | Chisq   | Df | Prob       |
|---------|------------------------|---------|----|------------|
| general | General association    | 64.7130 | 3  | 5.7775e-14 |
| rmeans  | Row mean scores differ | 1.2096  | 1  | 2.7140e-01 |
| cmeans  | Col mean scores differ | 64.7130 | 3  | 5.7775e-14 |
| cor     | Nonzero correlation    | 1.2096  | 1  | 2.7140e-01 |

```
$ALL
```

```
Cochran-Mantel-Haenszel Statistics for country by Rating  
Overall tests, controlling for all strata
```

|         | AltHypothesis          | Chisq  | Df | Prob       |
|---------|------------------------|--------|----|------------|
| general | General association    | 246.36 | 3  | 4.0051e-53 |
| rmeans  | Row mean scores differ | 91.317 | 1  | 1.224e-21  |
| cmeans  | Col mean scores differ | 246.36 | 3  | 4.0051e-53 |
| cor     | Nonzero correlation    | 91.317 | 1  | 1.224e-21  |

```
CMHtest(dat_cc,  
        types="ALL",  
        overall=T,  
        cscores="midrank",  
        strata="gender")
```

```
$`gender:Female`
```

```
Cochran-Mantel-Haenszel Statistics for country by Rating  
in stratum gender:Female
```

|         | AltHypothesis       | Chisq  | Df | Prob       |
|---------|---------------------|--------|----|------------|
| general | General association | 246.04 | 3  | 4.6999e-53 |

```

rmeans  Row mean scores differ 223.84  1 1.3178e-50
cmeans  Col mean scores differ 246.04  3 4.6999e-53
cor      Nonzero correlation 223.84  1 1.3178e-50

```

```
$`gender:Male`
```

```

Cochran-Mantel-Haenszel Statistics for country by Rating
  in stratum gender:Male

```

|         | AltHypothesis          | Chisq   | Df | Prob       |
|---------|------------------------|---------|----|------------|
| general | General association    | 64.7130 | 3  | 5.7775e-14 |
| rmeans  | Row mean scores differ | 8.4411  | 1  | 3.6683e-03 |
| cmeans  | Col mean scores differ | 64.7130 | 3  | 5.7775e-14 |
| cor     | Nonzero correlation    | 8.4411  | 1  | 3.6683e-03 |

```
$ALL
```

```

Cochran-Mantel-Haenszel Statistics for country by Rating
  Overall tests, controlling for all strata

```

|         | AltHypothesis          | Chisq  | Df | Prob       |
|---------|------------------------|--------|----|------------|
| general | General association    | 246.36 | 3  | 4.0051e-53 |
| rmeans  | Row mean scores differ | 149.89 | 1  | 1.8367e-34 |
| cmeans  | Col mean scores differ | 246.36 | 3  | 4.0051e-53 |
| cor     | Nonzero correlation    | 149.89 | 1  | 1.8367e-34 |

### 3.13.3 Answer

Using MODRIDIT scores, we get  $Q_{SMH} = 149.88, p < 0.001$ . Using TABLE scores, we get  $Q_{SMH} = 91.3, p < 0.001$ . There is sufficient evidence that there is a difference in the attitude toward school between the students from US and Korea.

- I would use the modridit scores because the spacing between ratings is non-uniform.
- You can use multiple scores and see whether the resulting inference is the same.
- The Somers' D (C|R) for females is 0.22 (0.19,0.25)
- The Somer's D (C|R) for males is 0.04 (0.01,0.07)

## 4 $s \times 2$ table

### 4.1 $s \times 2$ table

- Used when there are multiple groups defined by an ordinal variable with a dichotomous outcome.
- Example: Degree of gambling experience, degree of smoking experience, Educational Attainment

### 4.2 $s \times 2$ table

Assume  $X$  is an ordinal variable.

To account for ordinality, we assign a set of scores  $\mathbf{c}$  reflecting the relationship between response levels.

$$\mathbf{c} = \{c_1, c_2, c_3, \dots, c_s\} : c_1 < c_2 < \dots < c_s$$

In  $2 \times r$  tables,  $\mathbf{a} = (a_1, a_2)$  is effectively (0,1)

|       | $A_1$    | $A_2$    | Total    |
|-------|----------|----------|----------|
| $X_1$ | $n_{11}$ | $n_{12}$ | $n_{1+}$ |
| $X_2$ | $n_{21}$ | $n_{22}$ | $n_{2+}$ |
| ...   | ...      | ...      | ...      |
| $X_S$ | $n_{s1}$ | $n_{s2}$ | $n_{s+}$ |
| Total | $n_{+1}$ | $n_{+2}$ | $n$      |

### 4.3 $s \times 2$ table

The expected frequency can be expressed as

$$\bar{f} = \sum_{i=1}^s \sum_{j=1}^2 \frac{c_i a_j n_{ij}}{n}$$

#### 4.4 $s \times 2$ table

The null hypothesis  $H_0$ : There is no association between the two variables. To test this hypothesis, we can use a randomization statistic  $Q_{CS}$  such that

$$Q_{CS} = \frac{(\bar{f} - \mu_c \mu_a)^2}{v_c v_a / (n - 1)} = (n - 1) r_{ac}^2$$

##### **i** Note

- $\mu_c$  is the expected value of the row scores
- $\mu_a$  is the expected value of the column scores.
- $v_c$  is the variance of the row scores
- $v_a$  is variance of the column scores

##### **i** Note

$Q_{CS}$  is also referred to the correlation statistic.  $Q_{CS}$  follows a chi-square distribution with one degree of freedom.

##### **i** Note

$Q_{CS}$  is related to the Pearson correlation coefficient  $r_{ac}$

#### 4.5 Cochran-Armitage test

A similar statistic to the correlation statistic is the  $z^2$  statistic used in the Cochran-Armitage test.

- The Cochran-Armitage trend test tests for trends in binomial proportions across levels of an ordinal covariate.

$$z^2 = Q_{CS} \left[ \frac{n}{n - 1} \right]$$

#### 4.6 $s \times 2$ analysis: SAS

- The correlation statistic can be found in the SAS Output as the Mantel-Haenszel chi-square statistic from the `chisq` option.

- SAS can also perform the Cochran-Armitage trend test by including the **trend** option in the **tables** statement.
- The Pearson correlation coefficient and the Somers' D(C|R) are of particular interest in the **measures** output since both metrics account for ordinality.

```
proc freq;
weight count;
tables a*b/trend chisq measures;
run;
```

## 4.7 Example

A random sample of 200 married men, all retired, were classified according to education and number of children. The data is shown in the table.

- Is there sufficient evidence of association between the father's educational attainment and the number of children?
- Using the Cochran-Armitage trend test to determine if there is an increasing trend in number of children with increasing educational attainment.

```
dat <- expand.grid(education=c("secondary","college","graduate"), Kids = c("At most 3","More than 3"),
mutate(count=c(51,61,29,32,17,10))

xtabs(count ~ education+Kids ,data=dat) -> dat_cc
dat_cc
```

| education | Kids      |             |
|-----------|-----------|-------------|
|           | At most 3 | More than 3 |
| secondary | 51        | 32          |
| college   | 61        | 17          |
| graduate  | 29        | 10          |

## 4.8 Example

A random sample of 200 married men, all retired, were classified according to education and number of children. The data is shown in the table.

- Is there sufficient evidence of association between the father's educational attainment and the number of children?
- Using the Cochran-Armitage trend test to determine if there is an increasing trend in number of children with increasing educational attainment.

### 4.8.1 SAS Code

```
data educ;
input education $ kids3 $ count @@;
datalines;
secondary leq 51 secondary g 32
college leq 61 college g 17
graduate leq 29 graduate g 10
;
run;

proc freq data= educ order=data;
weight count;
tables education*kids3/chisq measures trend cl;
run;
```

①

① `chisq` gives  $Q_{CS}$ , `measures` provides Somers and pearson, `trend` gives cochrane-armitage, `cl` calculates confidence intervals.

### 4.8.2 Answer

$Q_{CS} = 3.47, p = 0.06$ . There is weak evidence of association between the father's educational attainment and number of offspring.

The one-sided Cochran-Armitage test yielded a p-value = 0.03, which is sufficient evidence to support a decreasing trend in number of children with increasing educational attainment.

## 5 Sets of $s \times 2$ tables

### 5.1 Sets of $s \times 2$ tables

- Just like 2x tables, there is also a statistic for the association of two ordinal variables for a combined set of strata called the extended Mantel-Haenszel correlation statistic ( $Q_{CSMH}$ ).
  - $Q_{CSMH}$  approximately follows a chi-square distribution with  $df=1$  when the combined strata sample sizes is greater than or equal to 40 (sufficiently large).

## 5.2 Example

The table contains two contingency tables of risk perception (minimal, moderate, and substantial) and adolescent tobacco usage by father's usage.

- Comment on whether there is an association between risk perception and adolescent usage when the father does not use tobacco.
- Is there a discernible trend in the proportions of adolescent usage over the levels of risk perception after controlling for father's tobacco usage?

**Table 4.5** Adolescent Smokeless Tobacco Usage

| Father's Usage | Risk Perception | Adolescent Usage |     | Total |
|----------------|-----------------|------------------|-----|-------|
|                |                 | No               | Yes |       |
| No             | Minimal         | 59               | 25  | 84    |
| No             | Moderate        | 169              | 29  | 198   |
| No             | Substantial     | 196              | 9   | 205   |
| Yes            | Minimal         | 11               | 8   | 19    |
| Yes            | Moderate        | 33               | 11  | 44    |
| Yes            | Substantial     | 22               | 2   | 24    |

## 5.3 Example

The table contains two contingency tables of risk perception (minimal, moderate, and substantial) and adolescent tobacco usage by father's usage.

- Comment on whether there is an association between risk perception and adolescent usage when the father does not use tobacco.
- Is there a discernible trend in the proportions of adolescent usage over the levels of risk perception after controlling for father's tobacco usage?

### 5.3.1 Contingency table

**Table 4.5** Adolescent Smokeless Tobacco Usage

| Father's Usage | Risk Perception | Adolescent Usage |     | Total |
|----------------|-----------------|------------------|-----|-------|
|                |                 | No               | Yes |       |
| No             | Minimal         | 59               | 25  | 84    |
| No             | Moderate        | 169              | 29  | 198   |
| No             | Substantial     | 196              | 9   | 205   |
| Yes            | Minimal         | 11               | 8   | 19    |
| Yes            | Moderate        | 33               | 11  | 44    |
| Yes            | Substantial     | 22               | 2   | 24    |

### 5.3.2 SAS Code

```
data tobacco;
length risk $11. ;
input f_usage $ risk $ usage $ count @@;
datalines;
no minimal no 59 no minimal yes 25
no moderate no 169 no moderate yes 29
no substantial no 196 no substantial yes 9
yes minimal no 11 yes minimal yes 8
yes moderate no 33 yes moderate yes 11
yes substantial no 22 yes substantial yes 2
;
proc freq;
weight count;
tables f_usage*risk*usage /cmh chisq measures trend;
tables f_usage*risk*usage /cmh scores=modridit;
run;
```

### 5.3.3 Answer

When the father is not a tobacco user,  $Q_{CS} = 34.3, p < 0.001$ , which means there is sufficient evidence of an association between risk perception and participant tobacco use.

$Q_{CSMH} = 39.3$  using modified ridit scores ( $Q_{CSMH} = 40.7$  for table scores), resulting to a p-value  $p < 0.001$ . There is sufficient evidence of an association between risk perception and participant tobacco use.

## 6 $s \times 2$ : Exact Analysis

### 6.1 Exact Analysis

- For small sample sizes,  $Q_{CS}$  does not satisfy the asymptotic requirements.
- We use exact tests to calculate the correct Mantel-Haenszel statistics.
- We manually calculate the Wilcoxon rank and logrank scores for each row.
  - Wilcoxon vs logrank: Logrank is better when there are more counts in the lower end of the row ordinal variable.

### 6.2 Exact Analysis: SAS

- The `exact` statement is used to perform an exact test on the correlation statistic.
  - `exact mhchi` would perform the statement on the Mantel-Haenszel statistic.
  - `scorout` outputs the scores used in the exact analysis

```
proc freq;  
tables a*b /scorout;  
exact mhchi;  
run;
```

### 6.3 Example

The table shows data from a study of mice who were exposed to a bacterial challenge and treated with the drugs carbenicillin or cefotaxmine (Bowdre, et al. 1983). Surviving mice were assessed at intervals of 6 to 24 hours. Is there evidence that there is a difference in performance between the two drugs?

### 6.3.1 Contingency Table

**Table 4.9** Mice Surviving Exposure to *Vibrio Vulnificus*

| Hours | Carbenicillin | Cefotaxime | Total | Ranks | Logranks |
|-------|---------------|------------|-------|-------|----------|
| 0–6   | 1             | 1          | 2     | 1.5   | 0.909    |
| 6–12  | 3             | 1          | 4     | 4.5   | 0.709    |
| 12–18 | 5             | 1          | 6     | 9.5   | 0.334    |
| 18–24 | 1             | 0          | 1     | 13    | 0.234    |
| 24–30 | 1             | 2          | 3     | 15    | –0.099   |
| 30–48 | 0             | 2          | 2     | 17.5  | –0.433   |
| 48–72 | 1             | 1          | 2     | 19.5  | –0.933   |
| 72–96 | 0             | 1          | 1     | 21    | –1.433   |
| > 96  | 0             | 1          | 1     | 22    | –2.433   |
| Total | 12            | 10         | 22    |       |          |

### 6.3.2 SAS Data

```
data mice;
input LogRank Treatment $ count @@;
datalines;
0.909 Ca 1 0.909 Ce 1
0.709 Ca 3 0.709 Ce 1
0.334 Ca 5 0.334 Ce 1
0.234 Ca 1 0.234 Ce 0
–0.099 Ca 1 –0.099 Ce 2
–0.433 Ca 0 –0.433 Ce 2
–0.933 Ca 1 –0.933 Ce 1
–1.433 Ca 0 –1.433 Ce 1
–2.433 Ca 0 –2.433 Ce 1
;
```

## 6.4 Example

The table shows data from a study of mice who were exposed to a bacterial challenge and treated with the drugs carbenicillin or cefotaxime (Bowdre, et al. 1983). Surviving mice were

assessed at intervals of 6 to 24 hours. Is there evidence that there is a difference in performance between the two drugs?

### 6.4.1 SAS Code

```
data mice;
input LogRank Treatment $ count @@;
datalines;
0.909 Ca 1 0.909 Ce 1
0.709 Ca 3 0.709 Ce 1
0.334 Ca 5 0.334 Ce 1
0.234 Ca 1 0.234 Ce 0
-0.099 Ca 1 -0.099 Ce 2
-0.433 Ca 0 -0.433 Ce 2
-0.933 Ca 1 -0.933 Ce 1
-1.433 Ca 0 -1.433 Ce 1
-2.433 Ca 0 -2.433 Ce 1
;

proc freq order=data;
weight count;
tables LogRank*Treatment / norow nocol nopct scorout chisq;
tables LogRank*Treatment / noprint scores=rank scorout chisq;
exact mhchi;
run;
```

- ① Using Logranks
- ② using ranks. Note that logranks preserves the order of the original data set.

### 6.4.2 Answer

The exact logrank analysis showed that  $Q_{CS} = 4.06, p = 0.04$ , which means we have sufficient evidence of an association/difference between the treatment and specimen survival.

However, the exact *rank* analysis showed that  $Q_{CS} = 3.51, p = 0.06$ , which means we have weak evidence of an association/difference between the treatment and specimen survival.

I would favor the logrank analysis better because it encapsulates the data better and accounts for the higher representation of the lower end of the ordinal variable.